FWF Österreichischer Wissenschaftsfonds

DMP: Prediction of Unemployment in Vienna Districts Using Tourism and Demographic Data

I General Information	
I.1 Administrative information	Principal Investigator: Nicolas Philipp Email Address: e12216446@student.tuwien.ac.at Project Title: Prediction of Unemployment in Vienna Districts Using Tourism and Demographic Data Institution: TU Wien DMP Version: 1.0 DMP Date: 06.04.2026
I.2 Data management responsibilities and resources	The principal investigator, Nicolas Philipp, is solely responsible for all data management activities within this project, including the implementation of this Data Management Plan and for ensuring it is reviewed and, if necessary, revised. Resource Requirements: • Storage: Both input datasets are publicly available via the Austrian Open Government Data portal and require no dedicated storage infrastructure • Software: All processing is performed using open-source Python libraries (pandas, scikit-learn), which incur no additional costs • Time: An estimated 6 hours is considered for data documentation and preparation of data for sharing The TU Wien Research Data Repository (https://test.researchdata.tuwien.ac.at/) is used for publication and incurs no additional costs.
II Data Characteristics	
II.1 Data description and collection or re-use of existing data	This project does not collect or produce new data. It exclusively re-uses two publicly available datasets from the Austrian Open Government Data portal (data.gv.at), both published by Stadt Wien under open government data licenses: 1. *Unemployed Persons Since 2002 - Districts of Vienna* (Stadt Wien - Wirtschaft und Finanzen, *Arbeitslos gemeldete Personen im Alter 15-64 Jahre nach Geschlecht (absolut und pro 1.000 EinwohnerInnen) im Jahresdurchschnitt seit 2002 - Bezirke Wien*, Open Government Data Austria, Dataset ID: 9462d680-ede9-40b9-8102-b9baebaa4fbb. Available at: https://www.data.gv.at/katalog/dataset/9462d680-ede9-40b9-8102-b9baebaa4fbb, 2025) • Format: CSV (75KB, 1586 rows)

	• Contains the features: NUTS (statistical region of Austria [always "AT13" for Vienna]), DISTRICT_CODE, SUB_DISTRICT_CODE, REF_YEAR (reference year), REF_DATE (reference date), SEX (0 = total, 1 = men, 2 = women), UEP_VALUE (average number of unemployed persons [age 15-64]), UEP_DENSITY (unemployed persons per 1,000 inhabitants) 2. *Guest Overnight Stays Since 2002 - Districts of Vienna* (Stadt Wien - Wirtschaft und Finanzen, *Gästeübernachtungen (absolut und pro 1.000 EinwohnerInnen) seit 2002 - Bezirke Wien*, Open Government Data Austria, Dataset ID: ae4ebf87-9f46-4f05-9e3d-dbff002b216d. Available at: https://www.data.gv.at/katalog/datasets/ae4ebf87-9f46-4f05-9e3d-dbff002b216d, 2025) • Format: CSV (38KB, 554 rows) • Contains the features: NUTS (statistical region of Austria [always "AT13" for Vienna]), DISTRICT_CODE, SUB_DISTRICT_CODE, REF_YEAR (reference year), REF_DATE (reference date), TOU_VALUE (total guest overnight stays), TOU_DENSITY (overnight stays per 1,000 inhabitants), POP_AVE (average population) Both datasets are openly licensed and freely accessible without restriction, intended for public reuse. Provenance will be documented by retaining the original dataset identifiers, source URLs, and publication years as listed above. The pre-processing steps: filtering (SEX = 0, DISTRICT_CODE ≠ 90000), removal of COVID-affected years (2020-2021), merging on DISTRICT_CODE and REF_YEAR, and one-hot encoding of the district code, will be fully documented in the Python source code (Jupyter Notebook, 282KB) via inline comments. The project generates multiple figures: 1. Distribution of unemployment values and tourism values per district as well as the unemployment density and tourism density p er district in terms of boxplots (PNG, 96KB) 2. Distribution of unemployment values and tourism values over time using boxplots (PNG, 48KB) 3. Scatterplot visualizing the best model's predictions (PNG, 68KB) The project exports the best performing model (random forest with the parameters: {'n_estimators': 200, 'max_depth': None, 'min_samples_split': 2}) as PKL file (6.81 MB) using the joblib library in Python. All formats chosen are open, non-proprietary, and do not require specialized or licensed software to read.
III Documentation and Data Quality	
III.1 Metadata and documentation	Metadata will be provided at both the dataset level (title, source institution, dataset IDs, URLs, temporal and geographic co verage) and the variable level (definitions for all fields used). A README.md will serve as the central documentation file, covering the aim of the project, references to the original datasets, reproduction instructions, pre-processing decisions, and results. A requirements.txt will document the Python environment for reproducibility. The repository will follow a simple folder structure separating datasets, source code, figures and models. Version control will be managed via Git.
III.2 Data quality control	The source datasets are published by an official government body (Stadt Wien) via the Austrian Open Government Data portal, p roviding an institutional baseline of quality assurance. Within the project, quality control is handled using the following pre-processing steps:

	• filtering (SEX = 0, DISTRICT_CODE ≠ 90000) • removal of COVID-affected years (2020-2021) • merging the datasets on DISTRICT_CODE and REF_YEAR • one-hot encoding of the district code All decisions regarding those measures are recorded in code comments.
IV Data Storage, Sharing, and Long-Term Preservation	
IV.1 Data storage and backup during the research process	Data and metadata will be stored in two locations during the research process: a local copy on the researcher's personal computer and a private GitHub repository, which provides automatic versioning and offsite backup. In the event of a technical incident, the full project can be restored from the GitHub repository. Since all input data is publicly available on the Austrian Open Government Data portal, it can be re-downloaded at any time, further reducing the risk of data loss. Access to the private repository is restricted to the principal investigator during active research. No sensitive or personal data is processed, so no special data protection measures are required beyond standard account security (password protection and two-factor authentication on GitHub).
IV.2 Data sharing and long-term preservation	Upon completion of the project, the Git repository will be made publicly available. Outputs will be published on the TU Wien Research Data Repository (https://test.researchdata.tuwien.ac.at/) which provides a persistent DOI and long-term archiving. The source datasets are maintained and updated by the city of Vienna at data.gv.at. The pre-processed dataset and outputs will be published under a CC BY 4.0 license, consistent with the open government data licenses of the source datasets and allowing broad reuse with attribution. Source code will be published under the MIT license. All tools needed to access and reuse the data (Python, pandas, scikit-learn) are open-source and freely available and listed in the requirements.txt file. The raw input files, processed dataset, source code, and key outputs (figures, model, README) will be preserved.
V Legal and Ethical Aspects	
V.1 Legal aspects	The input datasets are owned and published by Stadt Wien under open government data licenses, which explicitly permit free reuse including for research purposes. The derived outputs (processed dataset, source code, figures, model) are authored by Nicolas Philipp, who retains intellectual property rights over these contributions and will publish them under CC BY 4.0. No personal data is processed at any stage. Both source datasets consist entirely of aggregated statistics at the district and annual level. No individuals are identifiable from the data. GDPR therefore does not apply, and no consent, anonymization, or managed access procedures are required.
V.2 Ethical aspects	No significant ethical concerns are associated with this project. The data is publicly available, non-personal, and aggregated, and the research does not involve human participants, vulnerable groups, or sensitive topics. No ethical review by a committee is required. The project follows the principles of the European Code of Conduct for Research Integrity and the TU Wien Code of Ethics.